SPE-227507-MS

Assessing the Morphological and Mechanical Stability of PEEK in Severe Oilfield Conditions: An Investigation of High-Temperature HCl Acids and Supercritical CO_2 Exposure

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 DOI 10.2118/227507-MS

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

This research investigates the aging behavior of Polyetheretherketone (PEEK) when exposed to supercritical CO_2 ($scCO_2$) and acidic environments at elevated temperatures, conditions commonly encountered in the oil and gas industry, specifically downhole applications. With growing interest in replacing traditional carbon steel components, despite PEEK presents a promising alternative due to its superior chemical resistance, thermal stability, and mechanical properties in short term exposure to harsh environment, its aging response (long term exposure) is not well understood. To this end, PEEK was extensively aged at aggressive conditions (high temperature acid and sc-CO_2 and evaluated their impact incorporating both the physical, chemical and mechanical response for better understanding of PEEK compatibility for these particular conditions.

The study evaluates the long-term performance of PEEK under these harsh conditions through their impact on mechanical, physical and thermal properties. PEEK samples were exposed to HCl at 100°C and in $scCO_2$ at 150°C and 100 bar. After that, samples were analyzed thoroughly utilizing wide variety of characterization techniques to evaluate the uptake and its effect on morphology and mechanical response of PEEK. This study characterized PEEK through multiple analyses, including mechanical testing (hardness, tensile strength), chemical analysis via FTIR, and thermal analysis (DSC) to evaluate crystallinity and providing insights into its structure, properties, and behavior.

The findings demonstrate that PEEK offers significant stability when exposed to supercritical CO_2 and HCl at elevated temperatures. Effects of both environments on microstructure and chemistry are elucidated. Phenomena, such as secondary crystallization, are observed while very limited. Mechanical properties, such as elasto-plastic and fracture properties feature also limited modifications. This study reveals that exposure to HCl induces subtle plasticisation effects in PEEK, unlike $scCO_2$ which yields no such plasticisation effects. Importantly, both aging conditions trigger solely physical modifications, altering the material's

morphological, mechanical, and thermal characteristics without inducing significant chemical interaction that could lead to alter or generating new functional groups.

This research contributes valuable insights for the oil and gas industry, promoting the adoption of PEEK as a viable replacement for carbon steel in applications where long-term material resilience is critical, particularly in CO_2 transport, storage and well treatment processes.

Introduction

The integrity and durability of pipeline materials are crucial for the operation of Carbon Capture and Storage (CCS) and Enhanced Oil Recovery (EOR) ('Pandey & 'Tiwari, 2021; Sonke et al., 2022). Traditional metallic materials are prone to corrosion, particularly in downhole settings where aggressive media such as supercritical carbon dioxide ($scCO_2$) and hydrochloric acid (HCl) are prevalent(Nasr-El-Din et al., 2006; Nazarian et al., 2013). Non-metallic (NM) and composite pipes, such as reinforced thermoplastic pipe (RTP) and thermoplastic composite pipe (TCP), are being explored as alternatives(Badeghaish et al., 2019). Polyetheretherketone (PEEK) is a high-performance polymer used in harsh environments, but its interactions with $scCO_2$ and HCl under high-pressure, high-temperature (HPHT) conditions are not well understood (Badeghaish et al., 2024, 2025). As shown in Figure 1(a–c), HCl injection for stimulation and CO_2 transport pose challenges to composite or polymer liners due to acid and gas diffusion, leading to material degradation and mechanical property changes (Kermani et al., 2023; Sonke et al., 2022).

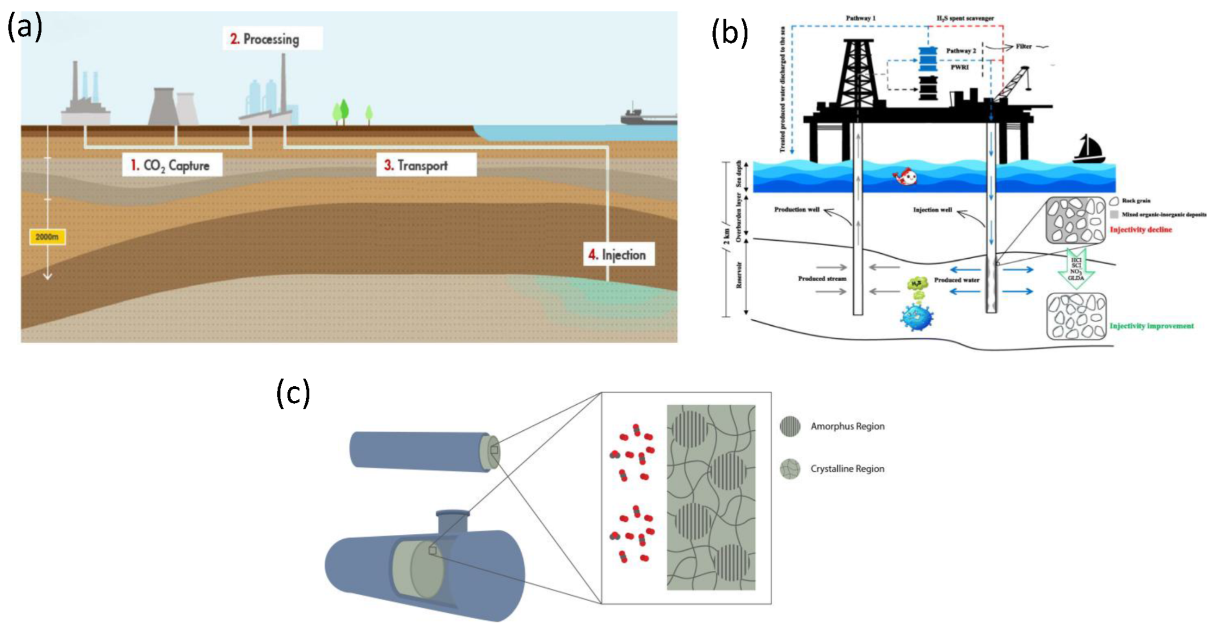

Figure 1—(a) Schematic illustration of the Carbon Capture and Storage (CCS) process (Sonke et al., 2022); (b) offshore PWRI to enhance well injectivity using HCl acidizing (Kermani et al., 2023); and (c) schematic for $scCO_2$ molecule diffusion challenges on polymer microstructure.

PEEK is a promising material for downhole applications due to its exceptional strength, thermal stability, and resistance to degradation (Badeghaish et al., 2024, 2025). The effects of acid on PEEK have been extensively studied in the literature, with a focus on understanding the material's behavior under various chemical and thermal environments. According to available studies, exposure to acids can alter the surface characteristics and microstructure of PEEK, leading to changes in its mechanical properties. For instance, sulfuric acid has been shown to introduce sulfonate functional groups, enhancing surface roughness and adhesion with dental adhesives, while hydrofluoric acid has a minimal effect on surface texture('Amornrat et al., 2022; Zhou et al., 2014). The literature also suggests that acid aging can influence the molecular mobility of PEEK, promoting structural rearrangement of its chains ('Amornrat et al., 2022; Zhou et al., 2014).

The use of PEEK for gas transportation and injection raises concerns about chemical compatibility, permeability, and resistance to rapid gas decompression. Literature suggests that $scCO_2$ significantly affects the thermal and rheological properties of polymers, including reducing viscosity and altering melting and glass-transition temperatures. Studies on PEEK have shown that $scCO_2$ exposure can alter its crystallinity, with some studies indicating that the plasticization of its amorphous phase can be reversed. For example, research has found that $scCO_2$ exposure can decrease the crystallization temperature of PEEK and alter its crystallization behavior, with the degree of change depending on the amount of CO_2 absorbed by the polymer (Mohammed et al., 2013; Wang et al., 2007). Additionally, studies have reported that the crystallinity of PEEK can increase noticeably at high temperatures, 140 °C, and that the effect of pressure on crystallinity is significant, with a notable rise in crystallinity observed under high pressures ('Bologna et al., 2007). However, there is a gap in research on the effects of $scCO_2$ on PEEK's performance at high temperatures and pressures, with existing studies highlighting complex interactions between temperature, pressure, and CO_2 solubility on PEEK's crystallinity. Further research is needed to fully understand the effects of $scCO_2$ on PEEK's properties and behavior, particularly in downhole applications where high temperatures and pressures are common. Further research is needed to fully understand the effects of $scCO_2$ on PEEK's properties and behavior, particularly in downhole applications where high temperatures and pressures are common. Other studies have demonstrated that the thermal aging at high temperatures can significantly affects PEEK'S mechanical properties with temperature above 250 °C causing molecular chain to oxide and break (Zhu et al., 2022).

This research aims to investigate the effects of $scCO_2$ and HCl on PEEK under HPHT conditions, focusing on understanding the interactions between PEEK and $scCO_2$, and the effects of HCl on PEEK's microstructure and mechanical properties. A comprehensive experimental study and was conducted to isolate the individual contributions of temperature, pressure, and CO_2 to PEEK's morphology and mechanical properties. Additionally, an evaluation of the morphology and mechanical properties of PEEK was conducted under HCl conditions to assess its performance and durability in such an environment.

Material and experiment

In this study, PEEK (semicrystalline thin film) samples with thicknesses of 50, 125, and 250 micrometers were tested. These samples were supplied by Victrex, UK, as part of the APTIV 1000 series, featuring a gloss/gloss surface finish. The same material was used for both types of aging tests.

PEEK thin film samples were aged in 37% aqueous HCl at two conditions: room temperature (25°C) and elevated temperature (100°C) for 336 hours. Samples were placed in a Teflon rack submerged in a glass container filled with HCl. Magnetic stirrers were used to maintain uniform acid flow across the samples as shown in in Figure (a). The acid uptake was evaluated after taking the samples out from acid.

In separate aging experiment, PEEK thin film samples were aged in supercritical CO_2 ($scCO_2$) at 150°C and 100 bar for 168 hours using a 500mL high-pressure stainless-steel cell as explained in Figure (b). CO_2 (99.9% purity) was introduced at 60bar and heated to reach $scCO_2$ conditions. After aging, the system was cooled and depressurized, and samples were sealed to prevent CO_2 loss before characterization. To isolate individual effects, three control conditions were tested for the same duration: T-Vac (150°C, vacuum), P-Ar (150°C, 100bar Ar), and T-CO_2 (150°C, 30bar CO_2). For both aging works, three sample types were used: square (25×25mm²) for morphology, rectangular (20×5mm) for Tg and modulus analysis, and dogbone (75×10mm) for tensile testing as shown in Figure 2 (c).

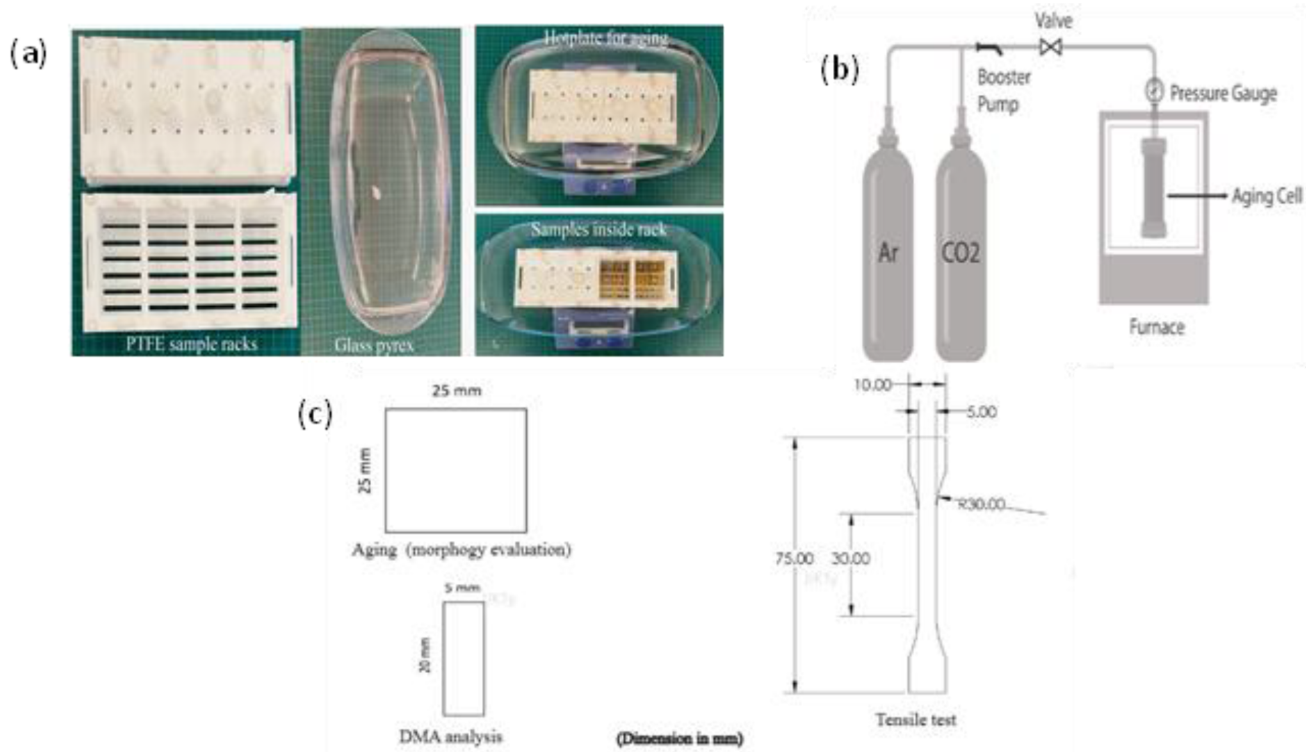

Figure 2—HCl aging: (a) experimental setup(Badeghaish et al., 2025); (b) scCO2 aging setup (Badeghaish et al., 2024); and (c) Samples specimen for mechanical testing (Badeghaish et al., 2024, 2025).

To assess diffusion behavior of acid and CO2, PEEK thin films were initially weighed, then immersed in aging environments. At specified intervals, samples were removed, washed, dried, and reweighed using a Mettler Toledo XP26 balance (±0.001 mg accuracy).

DSC tests were performed using a TA Q2000 to evaluate the crystallinity (Xc), glass transitions (Tg), and melting temperature (Tm) of PEEK before and after aging. Around 5 mg of each sample was sealed in a 40 µL aluminum pan and heated from 25°C to 400°C at 10°C/min under N_2 (50 mL/min). When a secondary melting enthalpy peak (ΔH_{m2} appears due to aging, it should be included in the crystallinity calculation. Crystallinity was calculated per ASTM D3418 as follows with $\Delta H_{m100\%}$ = 130 J/g for fully crystalline PEEK(Badeghaish et al., 2025).

$$X = \frac{\Delta H_{m1} + \Delta H_{m2}}{\Delta H_{m100\%}}$$

Attenuated Total Reflectance Fourier Transform Infrared (ATR-FTIR) spectra were recorded using a Thermo Scientific Nicolet iS10 spectrometer in the 600–4000 cm^{-1} range with a 4 cm^{-1} resolution to identify chemical bonds and functional group changes in PEEK samples before and after aging.

Tensile tests were conducted using a 5994 Instron machine (1 kN load cell, 1 mm/min) following ISO 527-2 standard, to evaluate changes in yield strength, ultimate strength, and elongation at break before and after aging. However, Nanoindentation tests were performed using a Nanotest Vantage machine with a Berkovich indenter, applying a 5 mN force with a 5-second dwell time, to determine the surface hardness and elastic modulus of PEEK samples before and after aging. The hardness and reduced elastic modulus of the samples were then calculated using the Oliver and Pharr method (Dahal et al., 2015; Voyiadjis et al., 2017), as follows:

$$H = \frac{\mathrm{Pm}}{\mathrm{A}}$$

where Pm is the maximum normal load and A is the projected contact area at the maximum load.

$$Er = \frac{\sqrt{\pi}}{2} S \frac{1}{\sqrt{A}}$$

$$\frac{1}{Er} = \frac{1-v2}{E} + \frac{1-vi^2}{Ei}$$

The reduced modulus of indentation contact, Er, is determined by several key parameters. These include S, the contact stiffness, which is calculated as dp/dh from the load-displacement curve as explained in Figure 3. Additionally, the elastic modulus of the indenter, Ei, is 1140 GPa, and the Poisson's ratio of the indenter, νi, is 0.07. The Poisson's ratio of the test sample, ν, is also an important factor, with a value of 0.41(Badeghaish et al., 2025; Jahanshahi et al., 2023).

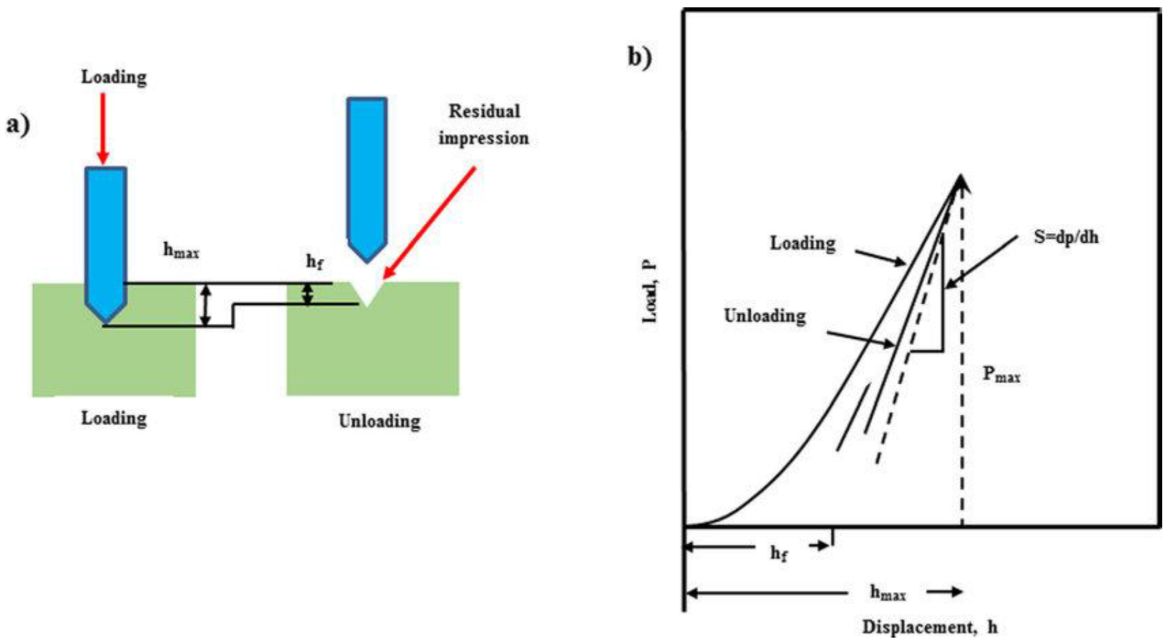

Figure 3—Nano-indentation schematics, illustrating (a) the working principle; and (b) a typical load-displacement curve, where the unloading slope represents contact stiffness (S) (Dahal et al., 2015).

Results and discussion

Effect of HCl on PEEK morphology and mechanical properties

The evolution of acid uptake with time was investigated. The thinnest samples (50 µm) reached maximum acid uptake with a 0.75% weight change after 48 hours at room temperature, while the thickest samples (250 µm) took 336 hours to reach a 0.65% weight change. The weight changes were relatively low due to PEEK's high crystallinity and low free volume, which makes it difficult for water/acid molecules to diffuse through the material. The low mass uptake of PEEK can be attributed to its tight molecular packing and low molecular mobility, which hinders the diffusion of water/acid molecules. Thicker samples showed lower maximum mass uptake due to the difficulty of acid diffusing into the bulk material. At higher temperatures (100°C), acid diffusion was faster, with all three thicknesses reaching saturation after 50 hours, due to the increased kinetic energy of acid molecules. This allows them to overcome energy barriers and diffuse through the material more easily(Badeghaish et al., 2024).

DSC analysis was perfumed to evaluate thermal properties of PEEK before and after aging in HCl. Evaluating the Tg, primary melting temperature (T_{m1}), secondary melting peak (T_{m2}) and degree of crystallinity helps us understand the thermal properties and structural changes of a material (Leisen et al., 2004). The Tg shows changes in the amorphous phase, indicating the flexibility of the material. The change in melting temperatures reveals changes in crystallinity, which affects the material's microstructure.

Figure 4 (a) shows the Tg of pristine PEEK was approximately 152.32°C. After aging in HCl at 100°C, the Tg decreased to 146.52°C and 146.25°C after 168h and 336h, respectively. This decrease is due to the plasticization effect caused by HCl diffusion, making the polymer more flexible. A small exothermic peak at 156°C indicated crystallization occurred due to acid interaction. Therefore, the results indicated that acid diffusion disrupts PEEK's crystalline and amorphous phases, leading to cold crystallization and a less stable crystalline structure compared to pristine PEEK (Badeghaish et al., 2024).

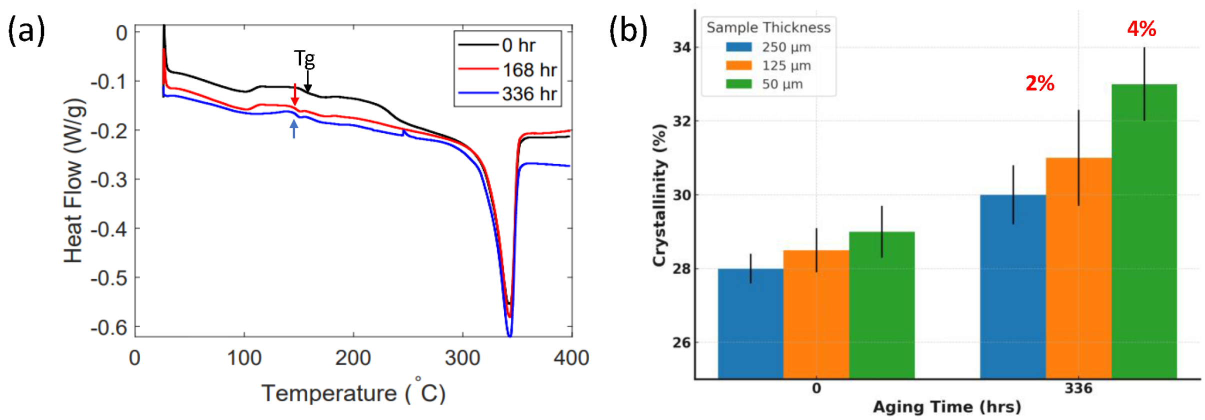

Figure 4—DSC analysis of PEEK aged in HCl at 100°C, showing (a) thermograms for 125µm thickness after various aging times and [14]; (b) Bar chart summary of crystallinity changes with respect to aging time for three different thicknesses.

Samples aged at 100°C showed a slight increase in crystallinity after 168 and 336 hours. The crystallinity increased by up to 2% for thicker samples (250 and 125 µm) and up to 4% for thinner samples (50 µm) after 336 hours in HCl as shown in Figure 4 (b). The interaction of acid induced and heat with the polymer caused a reorganization of the polymer chains, leading to a more ordered structure and increased crystallinity. The acid-induced interaction in PEEK is based on Lewis acid-base interactions. HCl donates a proton to the polymer chain, making it positively charged and forming hydrogen bonds between chains. This interaction occurs at the ether and ketone linkages, influencing the material's properties (Burwell, 1954; Duan et al., 2014; Kresge & Chiang, 1967).

The FTIR results for all thickness showed no significant changes in the PEEK spectra after exposure to HCl, indicating a weak interaction between PEEK and HCl. This weak interaction does not involve the formation or breaking of covalent bonds, and therefore, no new peaks appeared that would indicate an oxidation reaction. However, a closer look at the 1200-1800 cm-1 region revealed a slight change in the intensity of the carbonyl group peak at 1648 cm-1, suggesting a weak reaction between the ketone group in PEEK and the acid. Therefore, the interaction between PEEK and HCl is physical in nature, with no chemical reaction occurring. The FTIR spectrum of PEEK, as shown in the figure (5) for a specific thickness of 250 µm, exhibits the same functional groups as other thicknesses, indicating a consistent response (Alsar et al., 2021; Badeghaish et al., 2024).

The nanoindentation performed to evaluate the hardness and modulus of the surface of PEEK samples and this can be calculated using the Oliver and Pharr method as explained in method section Figure 3 (Dahal et al., 2015). The nanoindentation results showed that aging at 100°C had a significant effect on the hardness and modulus of the thinnest sample (50 µm thickness), with a 10% increase in hardness and 4% increase in modulus after 168 hours, and a 14% increase in hardness and 5% increase in modulus after 336 hours. This increase is attributed to the increased crystallinity of the sample. In contrast, thicker samples (125 and 250 µm) showed insignificant changes due to incomplete crystallization of the subsurface layer. The full crystallization of the thinnest sample, which was probed by the indentation, resulted in the observed increase in hardness and modulus, whereas the thicker samples' sublayers dominated the stress field, reducing the hardness increase (Badeghaish et al., 2024). Figure 6 shows a bar chart summary for modulus and hardness before and after aging in HCl at 100°C.

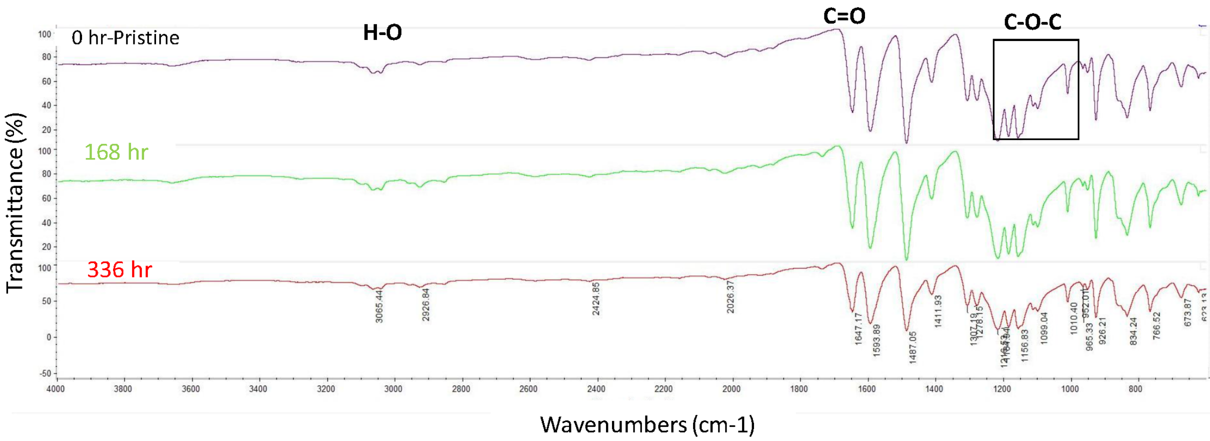

Figure 5—FTIR analysis of 250 μm PEEK samples aged in HCl at 100°C at 168 and 336 hr.

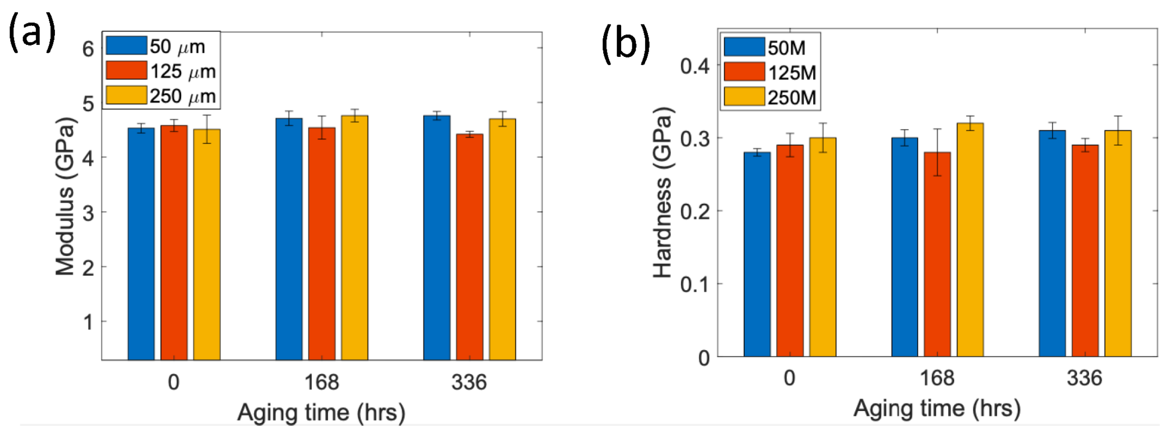

Figure 6—Nanoindentation bar chart summary of (a) Modulus; and (b) Hardness of PEEK samples aged in HCl at 100°C.

Effects of scCO2 on PEEK morphology and mechanical properties

Figure 7 (a) shows the changes in weight for PEEK samples with varying thicknesses under different aging conditions. The weight of the samples increased after aging due to gas absorption, with scCO_2 showing a notably larger increase in weight for thicker samples (125 and 250 μm), reaching a maximum weight change of 1.5%±0.07. This is because CO_2 gas was trapped inside the sample, resulting in a higher weight change. The weight change percentage of scCO_2 for thicker samples is higher than for thinner ones due to slower CO_2 desorption rates. Moreover, scCO_2 has a higher density (0.9 g/cm3) compared to Argon and regular CO_2 (0.0006-0.002 g/cm3). This higher density makes it more difficult for scCO_2 to desorb from materials especially after they have been heated. When scCO_2 is absorbed into PEEK, it can increase the material's weight. Figure 7(b) shows the insignificant effects of scCO_2 on the density of PEEK. The samples increase by almost 0.5% for samples with a thickness of 50 and 125 μm, and by 1% for samples with a thickness of 250 μm, due to the absorption of scCO_2 during aging. Moreover, the density of crystalline PEEK (1.40 g/cm3) is higher than amorphous PEEK (1.26 g/cm3) (Alsar et al., 2021). After aging, the crystallinity of PEEK increased by 9%, resulting in a higher density (Badeghaish et al., 2025).

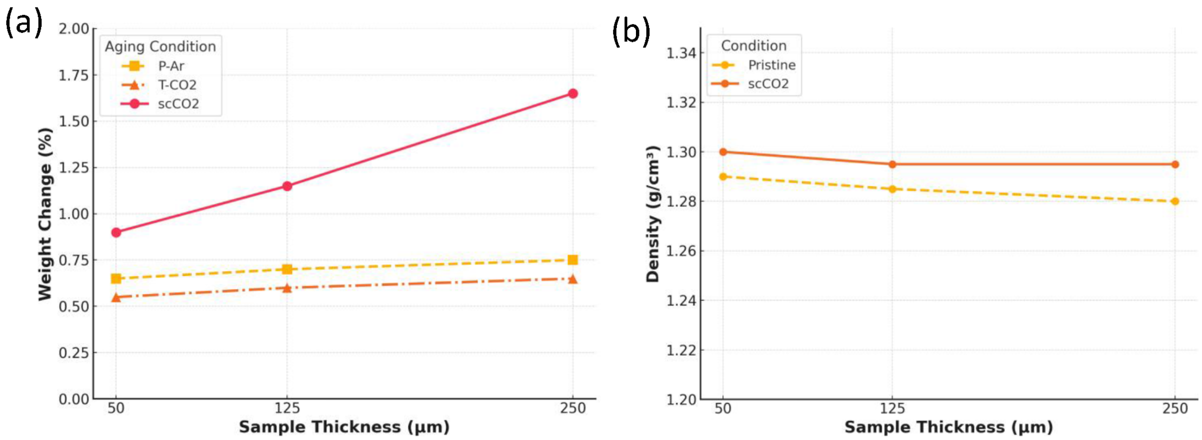

Figure 7—Evaluation of aged PEEK samples, showing (a) weight change percentage; and (b) density change.

DSC analysis was perfumed to evaluate thermal properties of PEEK before and after the aging in scCO2 and other gases conditions. Figure 8 (a) shows the DSC analysis of 250 µm pristine and aged PEEK samples under different conditions. A secondary crystal peak was observed at T_{m2} in all aged samples, indicating changes in microstructure. The T_{m2} increased with temperature and pressure, and was also influenced by the type of gas used. For example, samples aged in scCO2 had a higher Tm2 (198°C) compared to those aged in CO2 (184°C) or Ar. This suggests that scCO2's high diffusivity and solvent power enhanced the formation of more perfect and thermally stable crystals. Overall, the results indicate that the aging conditions had a significant impact on the crystallization behavior of the PEEK samples. All different samples thickness showed same response (Badeghaish et al., 2025).

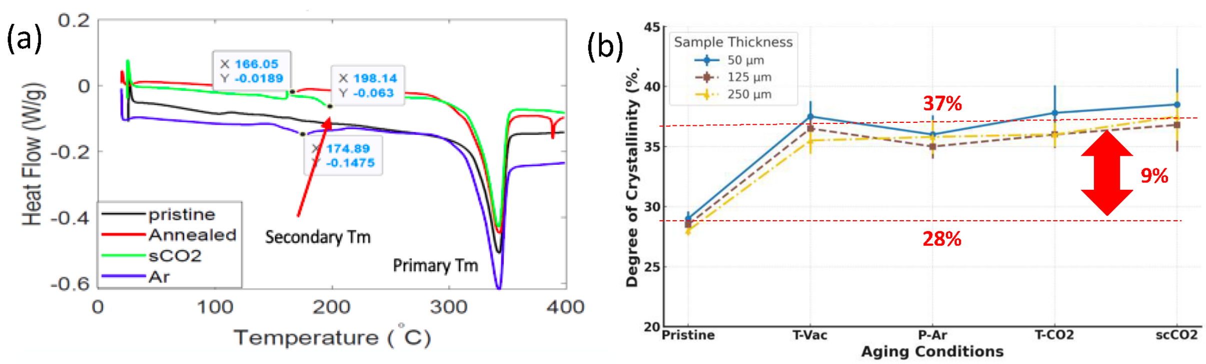

Figure 8—(a) DSC analysis of pristine and aged PEEK samples in various gas phases (scCO2, T-Vac, P-Ar, and T-CO2) at 150°C for 250 µm thickness [7]; and (b) evaluation of degree of crystallinity under different aging conditions.

The crystallinity of PEEK samples under different aging conditions is shown in Figure 8(b). The results indicate that unaged PEEK films have a consistent crystallinity of around 28.5%. In contrast, aged samples have a higher crystallinity of around 37%, regardless of the aging conditions. This suggests that temperature is the main factor driving crystallization (Badeghaish et al., 2025). Furthermore, aging increases crystallinity by enhancing both primary and secondary melting peaks (Lee & Kander, 1995; Wang et al., 2007).

FTIR analysis of PEEK samples performed. The results revealed no significant changes in peak intensities or new peaks, indicating that their chemical structures remained stable under various conditions. The results showed no evidence of chemical interactions, such as oxidation, and no peaks indicative of CO_2 absorption, suggesting that the aging process with $scCO_2$ did not alter the chemical composition of PEEK. The key functional groups, including ketone and ether groups, remained intact, confirming the stability of the PEEK structure (Badeghaish et al., 2025).

The nanoindentation results show that the load-depth response of pristine and aged PEEK samples of various thicknesses was similar under different aging conditions. The extracted hardness values and modulus reveal that aging led to a significant increase in hardness (10-17%) but had a negligible impact on the modulus. The hardness increase was more pronounced in samples aged under scCO2 and P-Ar conditions (16-17%) compared to annealed-Vac aging (10-13%) as shown in Figure 9(a). This suggests that temperature had a primary effect on hardness, while pressure and CO_2 diffusion had a secondary effect. Figure 9 (b) shows insignificant changes in modulus for all thickness. The increase in hardness was attributed to the transformation of amorphous regions into crystallites. Thinner samples showed a substantial increase in hardness (17%) due to crystallization caused by scCO2 diffusion from both sides, resulting in higher crystallinity. The stress field distribution for thinner samples extended up to 40 times the indentation depth, allowing the fully crystallized sample to be probed by indentation. In contrast, thicker samples had a stress field distribution dominated by sublayers of the indented surface that were not fully crystallized, resulting in a smaller increase in hardness. These findings support the hypothesis that diffusion in PEEK is primarily a surface phenomenon (Badeghaish et al., 2025).

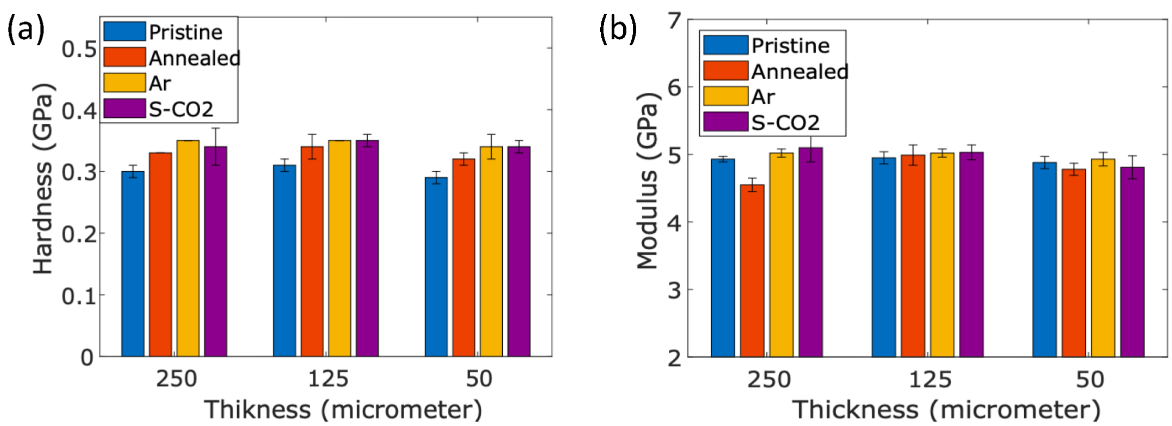

Figure 9—Nanoindentation bar chart summary of (a) Hardness, and (b) Modulus of PEEK samples of pristine and aging at different conditions.

The stress-strain response of 250-μm-thick PEEK samples aged under different conditions is shown in Figure 10. The curve exhibits a characteristic shape, with an initial linear region representing the elastic deformation of the material, followed by a non-linear region indicating plastic deformation, and finally, a rapid drop in stress signifying material failure. The ultimate and yield strengths of the aged samples increased, while the failure strain decreased, indicating a tendency towards brittleness due to aging. Specifically, the yield and ultimate strengths of $scCO_2$-aged PEEK samples were 25% and 8% higher than those of pristine samples, respectively. This enhancement in mechanical properties can be attributed to the increase in crystallinity, which leads to a more ordered molecular structure and enhanced chain interactions. The proportionality between mechanical properties and degree of crystallinity is well-established, with increased crystallinity resulting in higher strength and modulus of elasticity (Mohammed et al., 2013). The notable increase in yield strength of $scCO_2$-aged samples can be attributed to the reduced polymer chain mobility in the amorphous regions, resulting from denser packing and enhanced chain interactions. Conversely, the reduction in strain at failure under all aging conditions can be attributed to the enhanced crystallinity, which deteriorates the ability of the amorphous phase to deform and increases brittleness (Badeghaish et al., 2025).

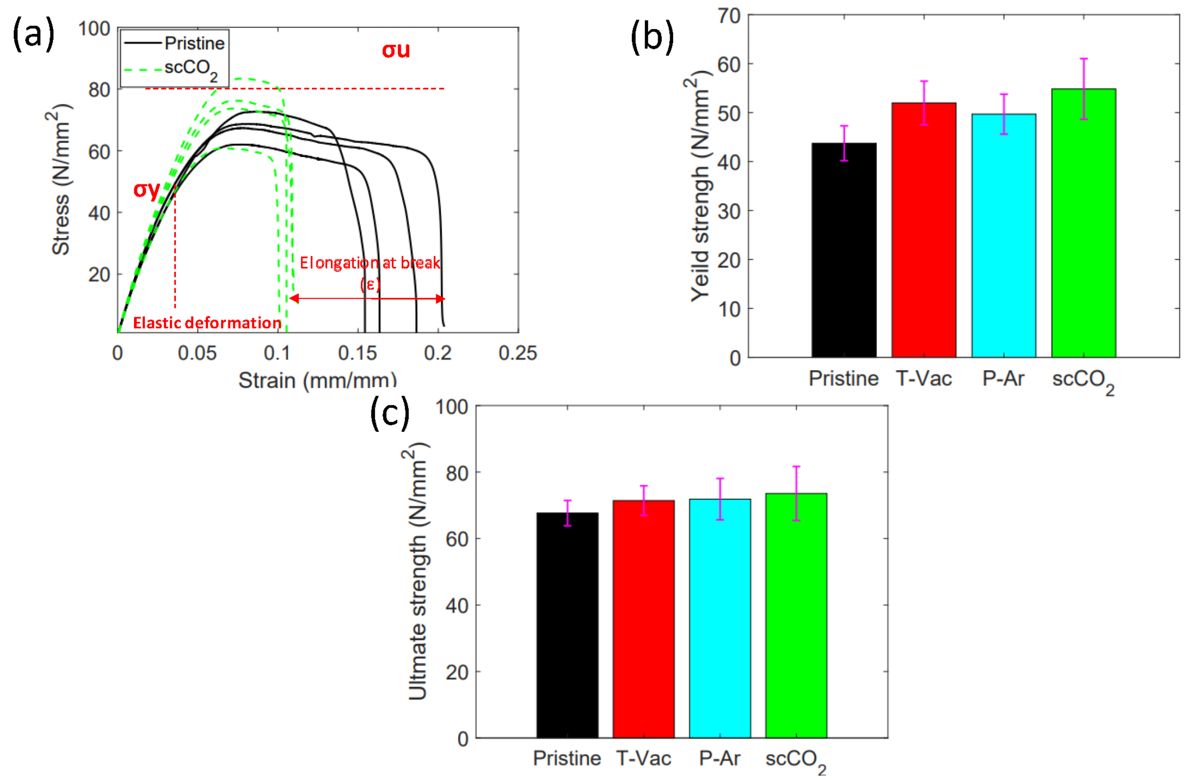

Figure 10—(a) Stress-Strain curves of 250 µm PEEK samples under $scCO_2$ aging conditions; (b) summary chart of yield strength; and (c) ultimate strength trend for all aging conditions.

Conclusion

The evaluation of PEEK samples exposed to HCl at 100°C revealed that acid diffusion leads to a decrease in Tg and an increase in crystallinity due to acid induced crystallinity and temperature effects. The interaction between PEEK and HCl is physical in nature, with no chemical reaction occurring. Nanoindentation results showed that aging at 100°C increases the hardness and modulus of thinner samples due to increased crystallinity. The findings show that PEEK's properties are slightly affected by acid diffusion, leading to changes in its thermal and mechanical behavior.

This study also reveals that $scCO_2$ aging induces physical changes in PEEK, without any chemical interactions, leading to a 9% increase in crystallinity. The absence of chemical reactions is confirmed by FTIR analysis, which shows no new peaks or significant changes in peak intensities. The increased crystallinity is accompanied by a shift in the T_{m2} at 198°C, indicating a change in the material's microstructure compared to others conditions (Ar, CO_2 and Vac-T). This shift in T_{m2} suggests that the $scCO_2$ aging process enables the formation of more perfect and thermally stable crystals, resulting in enhanced physical properties. The increased crystallinity and shift in T_{m2} are attributed to the physical interactions between $scCO_2$ and PEEK, which facilitate the reorganization of the polymer chains and lead to a more ordered crystal structure. The water contact angle also increases due to surface roughness and crystallinity. The nanoindentation results showed a significant increase in hardness (10-17%) after $scCO_2$ aging, with a more pronounced increase in thinner samples. Additionally, the stress-strain response shows increased ultimate and yield strengths, but decreased failure strain, indicating brittleness.

References

Alsar, Zh., Rakhadilov, B., Almasov, N., Serik, N., Spitas, C., Kostas, K., & Insepov, Z. (2021). Physical-chemical characterization of a PEEK surface irradiated with electron beams.

Amornrat, M., Chaiyasit, B., Pisaisit, C., Patcharawan, S. (2022). The effect of surface pre-treatments and dental adhesives on shear bond strength in polyetheretherketone (PEEK). *Journal of International Dental and Medical Research,*, 1503–1510.

Badeghaish, W., Noui-Mehidi, M., & Salazar, O. (2019, October 21). The Future of Nonmetallic Composite Materials in Upstream Applications. SPE Gas & Oil Technology Showcase and Conference. https://doi.org/10.2118/198572-MS

Badeghaish, W., Wagih, A., Colin, X., Hoteit, H., & Lubineau, G. (2025). Effect of supercritical CO2 aging on the microstructure and mechanical properties of PEEK. *Polymer Degradation and Stability*, **233**, 111155. https://doi.org/10.1016/J.POLYMDEGRADSTAB.2024.111155

Badeghaish, W., Wagih, A., Rastogi, S., & Lubineau, G. (2024). Effect of high-temperature acid aging on microstructure and mechanical properties of PEEK. *Polymer Testing*, **134**, 108429. https://doi.org/10.1016/J.POLYMERTESTING.2024.108429

Bologna, S., Del Re, G., Mascia, L., & Spagnoli, G. (2007). Crystallization of PC and PEEK with supercritical carbon dioxide. *IChea Ischia, Italy*, 24–27.

Burwell, R. L. (1954). The Cleavage of Ethers. *Chemical Reviews*, **54**(4), 615–685. https://doi.org/10.1021/cr60170a003

Dahal, T., Gahlawat, S., Jie, Q., Dahal, K., Lan, Y., White, K., & Ren, Z. (2015). Thermoelectric and mechanical properties on misch metal filled p -type skutterudites Mm0.9Fe4-xCoxSb12. *Journal of Applied Physics*, **117**(5). https://doi.org/10.1063/1.4906954

Duan, Y., Bai, R., Tian, J., Chen, L., & Yan, X. (2014). Hydrogenation of Aldehydes and Ketones to Corresponding Alcohols with Methylamine Borane in Neat Water. *Synthetic Communications*, **44**(17), 2555–2564. https://doi.org/10.1080/00397911.2014.909489

Jahanshahi, M., Mofidian, R., Hosseini, S. S., & Miansari, M. (2023). Investigation of mechanical properties of granular γ-alumina using experimental nano indentation and nano scratch tests. *SN Applied Sciences*, **5**(6), 164. https://doi.org/10.1007/s42452-023-05388-7

Kermani, H. M., Bonto, M., & Nick, H. M. (2023). Chemical solutions for restoring scaling-induced injectivity impairment in offshore produced water re-injection. *Science of The Total Environment*, **903**, 166597. https://doi.org/10.1016/J.SCITOTENV.2023.166597

Kresge, A. J., & Chiang, Y. (1967). The hydrolysis of ethyl vinyl ether. Part I. Reaction mechanism. *Journal of the Chemical Society B: Physical Organic*, 53. https://doi.org/10.1039/j29670000053

Lee, J. R., & Kander, R. G. (1995). The Application of Supercritical Fluid Technology to High-Performance Polymers (pp. 131–139). https://doi.org/10.1021/bk-1995-0603.ch008

Leisen, J., Beckham, H. W., & Sharaf, M. A. (2004). Evolution of Crystallinity, Chain Mobility, and Crystallite Size during Polymer Crystallization. *Macromolecules*, **37**(21), 8028–8034. https://doi.org/10.1021/ma0489974

Mohammed, M. H., Banks, W. M., Hayward, D., Liggat, J. J., Pethrick, R. A., & Thomson, B. (2013). Physical properties of poly(ether ether ketone) exposed to simulated severe oilfield service conditions. *Polymer Degradation and Stability*, **98**(6), 1264–1270. https://doi.org/10.1016/J.POLYMDEGRADSTAB.2013.02.014

Nasr-El-Din, H. A., Al-Mohammad, A. M., Al-Shurei, A. A., Merwat, N. K., Erbil, M. M., & Samuel, M. (2006). Restoring the injectivity of water disposal wells using a viscoelastic surfactant-based acid. *Journal of Petroleum Science and Engineering*, **54**(1–2), 10–24. https://doi.org/10.1016/J.PETROL.2006.07.001

Nazarian, B., Held, R., Høier, L., & Ringrose, P. (2013). Reservoir Management of CO2 Injection: Pressure Control and Capacity Enhancement. *Energy Procedia*, **37**, 4533–4543. https://doi.org/10.1016/J.EGYPRO.2013.06.360

Pandey, L., & Tiwari, P. (2021). Microbial Enhanced Oil Recovery: Principles and Potential.

Sonke, J., Bos, W. M., & Paterson, S. J. (2022). Materials challenges with CO2 transport and injection for carbon capture and storage. *International Journal of Greenhouse Gas Control*, **114**, 103601. https://doi.org/10.1016/J.IJGGC.2022.103601

Voyiadjis, G. Z., Samadi-Dooki, A., & Malekmotiei, L. (2017). Nanoindentation of high performance semicrystalline polymers: A case study on PEEK. *Polymer Testing*, **61**, 57–64. https://doi.org/10.1016/J.POLYMERTESTING.2017.05.005

Wang, D., Gao, H., Jiang, W., & Jiang, Z. (2007). Effect of supercritical carbon dioxide on the crystallization behavior of poly(ether ether ketone). *Journal of Polymer Science Part B: Polymer Physics*, **45**(21), 2927–2936. https://doi.org/10.1002/polb.21150

Zhou, L., Qian, Y., Zhu, Y., Liu, H., Gan, K., & Guo, J. (2014). The effect of different surface treatments on the bond strength of PEEK composite materials. *Dental Materials*, **30**(8), e209–e215. https://doi.org/10.1016/J.DENTAL.2014.03.011

Zhu, C., Zhang, H., & Li, J. (2022). Thermal aging study of PEEK for nuclear power plant containment dome. *Journal of Polymer Research*, **29**(1), 5. https://doi.org/10.1007/s10965-021-02839-w